

1. Draw an evaluation graph for the configuration $\langle S, \sigma \rangle$ where $S \equiv [x := 1 \parallel x := -1]; y := y + x$.
2. Draw an evaluation graph for the configuration $\langle W, \{x = 0, y = 1, n = 2\} \rangle$ where $W \equiv \mathbf{while} \ x < n \ \mathbf{do} \ [x := x + 1 \parallel y := y * 2] \ \mathbf{od}$.
3. Given the following two threads written in provable triples. Answer the following questions and justify your answers.

$$\begin{aligned} S_1 &\equiv \{z = x = 0\} \ y := x + 2 \ \{y = 2\} \\ S_2 &\equiv \{x \leq 0\} \ z := 1 \ \{z \geq x\} \end{aligned}$$

- a. Are these two threads disjoint?
 - b. Do they have disjoint conditions?
 - c. Find possible pre- and post-conditions and create a full proof outline under total correctness for the parallel statement $[S_1 \parallel S_2]$ using the Parallelism rule.
4. Create a Hilbert style proof for the following triple under total correctness using the Sequentialization rule.

$$\{z = x = 0\} [y := x + 2 \parallel z := 0] \{y = 2 \wedge z \geq x\}$$
 5. Answer the following questions about the two following threads written in proof outlines. Here, T_1, T_2 and T_3 are statements. Remember, we only need to check if atomic statements cause interference.

$$S_1^* \equiv \{p_1\} \ \mathbf{if} \ B_1 \ \mathbf{then} \ \{p_2\} < T_1 > \ \mathbf{else} \ \{p_3\} \ \mathbf{skip} \ \mathbf{fi} \ \{p_4\}$$

$$S_2^* \equiv \{q_1\} < T_2 >; \ \{\mathbf{inv} \ q_2\} \ \mathbf{while} \ B_2 \ \mathbf{do} \ \{q_3\} < T_3 > \ \mathbf{od} \ \{q_4\}$$

- a. List the interference freedom checks to decide whether S_1^* interferes with S_2^* .
 - b. List the interference freedom checks to decide whether S_2^* interferes with S_1^* .
6. Find possible pre- and post-conditions and create a full proof outline under total correctness for the following partial proof outline.

$$\{p\} [S_1^* \parallel S_2^*] \{q\}$$

Here

$$S_1^* \equiv \{x \geq 0\} < x := x + 1; y := x + 2 > \ \{y \geq 3\}$$

$$S_2^* \equiv \{z \geq 3\} \ x := z; \ \{x \geq 3\} \ y := x \ \{y \geq 3\}$$

7. Create a Hilbert style formal proof for the following triple under total correctness.

$$\{k > 0\} \ \mathbf{await} \ x = 1 \ \mathbf{then} \ y := 0 \ \mathbf{end} \ \{y < k\}$$

8. Given the following proof outlines.

$$S_1^* \equiv \{x = 1\} \ y := x + k; \ \{y = k + 1 \wedge x = 1\} \ \mathbf{wait} \ y = 0 \ \{x = 1\}$$

$$S_2^* \equiv \{k > 0\} \ \mathbf{await} \ x = 1 \ \mathbf{then} \ y := 0 \ \mathbf{end} \ \{y = 0 \wedge y < k\}$$

Prove that $\{\dots\} [S_1^* \parallel S_2^*] \{\dots\}$ is deadlock-free.